School of Information Technology, Engineering, Mathematics and Physics

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CS230: Requirements Engineering
Semester 1, 2022
Mode: Face to Face

Short Test 1
Date: Friday 29 April 2022, Time: 12.00 – 12.50pm (FJT)

Time Allowed: 50mins
30 marks (Weight: 7.5%)

Student ID: _____
Student Name: _____

LEARNING OUTCOMES:

- Identify various requirement elicitation techniques for software development.

INSTRUCTIONS

1) This exam has two sections:
   a) Section A: Short-answer questions (20 marks)
   b) Section B: Case Study (10 marks)
2) Answer all questions in this booklet. All questions are compulsory.
3) The total number of pages including this cover sheet is 6. Extra answer sheet is also attached.

~ALL THE BEST~

Section A: Short Answer Questions *(20 Marks)*

Question 1 (3 marks)

Answer the question based on the types of requirements.

a) Differentiate between Information Acquisition and Requirements Elicitation? (1 mark)

b) What are user requirements and system requirements and for whom they are written? (2 marks)

Question 2 (3 marks)

Classify the following requirements into functional and non-functional requirements. *(Write functional and non-functional beside the requirement in the space provided)*

a) A system user shall be able to search the vaccination record for specific person. _____

b) The system shall generate vaccination records, vaccination percentage, a list of vaccinated people on daily, weekly or monthly basis. _____

c) The system should be able to protect data from intruders and hash all passwords. _____

d) The mobile app should load in 3 seconds when the user opens the app. _____

e) The system shall verify that the user's entered credentials are correct. _____

f) The system should be follow style guide or Ministry, responsive and easy to use by vaccination staff. _____

Question 3 (2 marks)

Explain why a combination of ethnography and prototyping is useful for requirements elicitation?

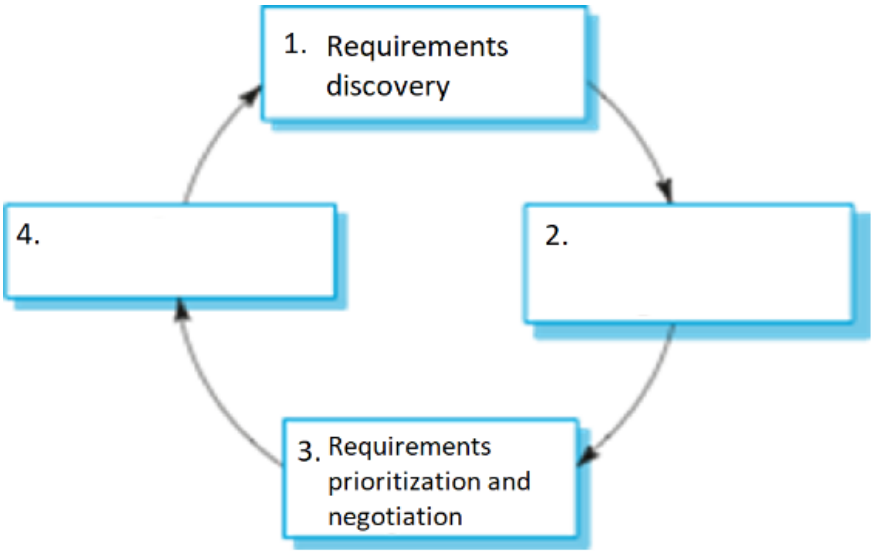
Question 4 (3 marks)

A software company is working on the Food ordering and delivery mobile app development for a Restaurant. Sera is tasked to collect all relevant information about how things are currently done, list and explain the three fact finding techniques she can use for this.

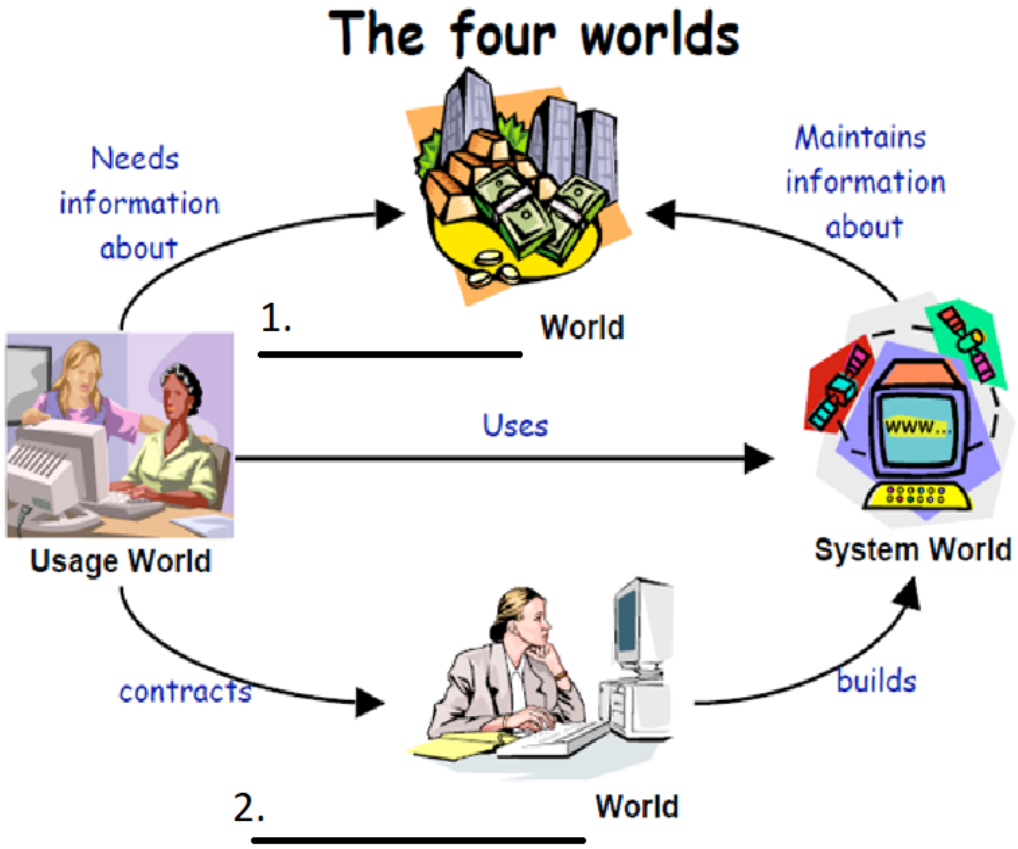
Question 5 (5 marks)

Fill in the blanks.

a) Write the missing Requirements Elicitation and Analysis Process at 2 and 4. (2 marks)



b) Write the missing names of the Four Worlds in Information Acquisition at the place marked 1 and 2 in the figure. With examples briefly explain the Usage and System world in Four Worlds. (2 marks)



c) With examples briefly explain the Usage World in Four Worlds. (1 mark)

Question 6 **(4 marks)**

a) Give two reasons why eliciting requirements are difficult? (2 marks)

Peter's company has signed a deal to develop a new Inventory system for a supermarket. One of the specific elicitation techniques is Interviews. He would like to interview requirement elicitation.

b) List and explain two things Peter must do to ensure that he gets all of the right information during the interview session.? (2 marks)

Section B: Case Study *(10 Marks)*

Covid-19 Vaccination System

The purpose of the Covid-19 Vaccination System is to provide online registration to citizens and vaccination records for the Ministry of Health. Firstly, the eligible citizens fills out a vaccination application with their birth registration information and vaccination site. After receiving the application the system sends the information to the Health officials at the site who will administer the vaccination and the Ministry of Health (MOH) for verification. The Ministry of Health checks and verifies all the information provided. If everything is fine then MOH gives approval which is saved in the system and a notification is sent to the Health officials at the vaccination site to check the available dates for the vaccination. A copy of the notification is also sent to the eligible citizens. The Health official then sends the available dates to the user via the system. If the user accepts a certain date and gets vaccinated. Upon vaccinated, vaccination detail for the citizen is entered into the system by the Doctor who gave the vaccination and the information is now stored. The vaccination certificate and card are generated and sent to the user. A daily report on vaccination is sent to the Ministry of Health and local Police department with vaccination percentage, and the number of vaccinated people in the area. The citizens are required to print a copy of the certificate and card and produce on-demand where needed.

Question 7 (2 mark)

List any two functional requirements of the Covid-19 Vaccination System.

Question 8 (8 marks)

Create a Use Case Diagram for the Covid-19 Vaccination System case study.

THE END

EXTRA ANSWER SHEET – USE ONLY WHEN NEEDED

- Mark the question number properly if used.